

Logical Operators - Zero to Hero Questions List

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0.1 Logical Operators (&&, ||, !) – Practice Questions in C

0.1.1 Beginner Level

1. Check if a number is positive and even. **Hint:** Use &&.
 2. Check if a number is between 1 and 100. **Hint:** Two conditions with &&.
 3. Check if a character is a lowercase letter. **Hint:** Use a range check with &&.
 4. Check if a number is divisible by both 2 and 5. **Hint:** Combine two modulus operations with &&.
 5. Check if a person is eligible to vote (age ≥ 18). **Hint:** Use a logical expression.
 6. Check if a number is not zero. **Hint:** Use !.
 7. Check if a number is either positive or zero. **Hint:** Use ||.
 8. Check whether a character is a vowel (a,e,i,o,u). **Hint:** Multiple conditions with ||.
 9. Check if a number is outside the range 10–20. **Hint:** Use ! with a range condition.
 10. Check if a number is either divisible by 3 or by 7. **Hint:** Use ||.
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0.1.2 Easy-Intermediate Level

11. Check if a year is a leap year candidate (divisible by 4). **Hint:** Use % and logical operators.
12. Check if a character is an alphabet letter. **Hint:** Two ranges joined by ||.
13. Check if a number is a two-digit positive number. **Hint:** Use &&.
14. Check if a number is not positive. **Hint:** Use !.
15. Check if a student passed both subjects. **Hint:** Marks > passing marks in both subjects.
16. Check if a number is neither divisible by 2 nor by 3. **Hint:** Use ! with ||.
17. Check if temperature is within a safe operating range. **Hint:** Lower and upper limits with &&.
18. Check whether a character is a digit. **Hint:** ASCII range check.
19. Check if at least one of two numbers is negative. **Hint:** Use ||.
20. Check if both numbers are non-zero. **Hint:** Use &&.

0.1.3 Intermediate Level

21. Check if a number lies in either range 1–10 or 20–30. **Hint:** Combine two range checks with `||`.
 22. Check if exactly one of two numbers is positive. **Hint:** One true and one false condition.
 23. Check if a character is not an uppercase letter. **Hint:** Use `!`.
 24. Check if a person qualifies for a discount (senior citizen or student). **Hint:** Use `||`.
 25. Check if a number is positive but not even. **Hint:** `&&` and `!`.
 26. Check if three numbers are all positive. **Hint:** Chain multiple `&&`.
 27. Check if at least one of three numbers is zero. **Hint:** Chain multiple `||`.
 28. Check whether a character is a consonant. **Hint:** Alphabet check + NOT vowel.
 29. Check if a number is not in the range 50–100. **Hint:** Negate a range condition.
 30. Check if a user entered either ‘Y’ or ‘y’. **Hint:** Use `||`.
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0.1.4 Advanced Level

31. Check if a year is a leap year using the complete rule. **Hint:** Combine `&&` and `||`.
 32. Check if all three sides can form a valid triangle. **Hint:** Multiple `&&` conditions.
 33. Check if exactly one of three numbers is positive. **Hint:** Count logical truths.
 34. Check if a character is a valid hexadecimal digit. **Hint:** Numeric range OR alphabetic range.
 35. Check if a number belongs to neither of two ranges. **Hint:** Use `!`.
 36. Check if an employee qualifies for promotion (experience and performance criteria). **Hint:** Multiple `&&`.
 37. Check if at least two out of three conditions are true. **Hint:** Use combinations of `&&` and `||`.
 38. Check if a password meets basic conditions (length, digit, etc.). **Hint:** Multiple logical expressions.
 39. Check whether a point lies inside a rectangle. **Hint:** Coordinate range checks.
 40. Check if a number is either a multiple of 4 or 6 but not both. **Hint:** Exclusive condition.
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0.1.5 Challenge Level

41. Check if exactly two of three numbers are even. **Hint:** Evaluate all combinations.
42. Check if exactly one of four conditions is true. **Hint:** Logical counting.

43. Check whether a date is potentially valid (day and month ranges). **Hint:** Multiple range checks.
 44. Check if a character is a printable ASCII character. **Hint:** ASCII range.
 45. Check whether a point lies in the first or third quadrant. **Hint:** Coordinate signs.
 46. Check if a number is positive, even, and a multiple of 5. **Hint:** Three `&&` conditions.
 47. Check if a number is negative or greater than 100. **Hint:** Use `||`.
 48. Check if none of three entered numbers are zero. **Hint:** Multiple `&&`.
 49. Check if all entered characters are vowels. **Hint:** Reuse vowel logic with `&&`.
 50. Check whether exactly one of two conditions is true (Logical XOR). **Hint:** `(A && !B) || (!A && B)`.
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0.1.6 Master Challenge Questions

51. Admission eligibility checker.
52. Loan approval checker.
53. Scholarship eligibility checker.
54. Smart home alarm activation logic.
55. ATM withdrawal validation.
56. Traffic signal decision simulator.
57. Employee attendance validation.
58. University course registration eligibility.
59. Online exam access validation.
60. Multi-condition game level unlock system.

Hint for 51–60: Mix `&&`, `||`, and `!` extensively in a single logical expression.